

Appendix A: Illustrations and Conceptual Schematics

Figure 1: The Thick Brane Interface (Σ)

A cross-sectional view of our 4D manifold showing finite topological ‘thickness’ within the 5th and 6th dimensions, illustrating gauge boson (photon) localization vs. graviton permeability.

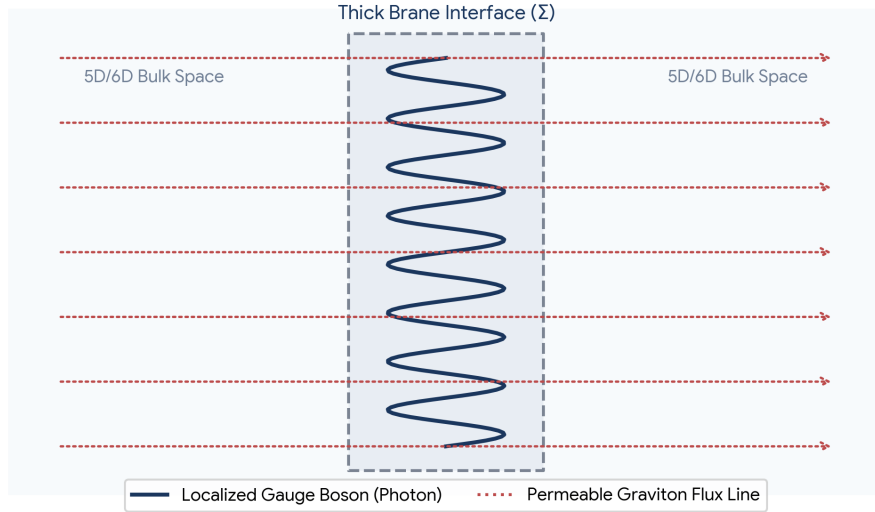


Figure 2: The Screening Gate (g_{eff})

A plot of the gate function $g_{\text{eff}}(R)$ vs. local curvature (R). Shows the sharp transition from 0 (high-density/local) to 1 (low-density/voids), representing the suppression of bulk coupling in local gravity environments.

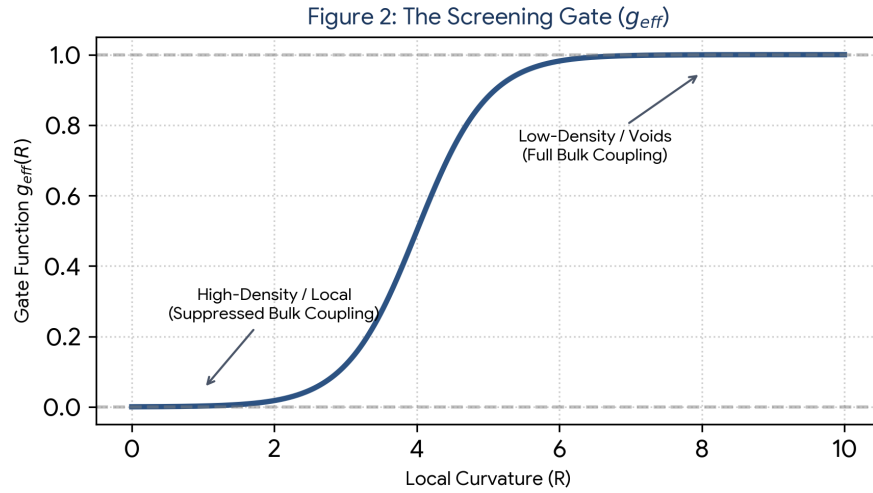


Figure 3: Evanescent Field Decay

A mathematical plot showing the amplitude of a gauge field $A_\mu(y)$ decaying exponentially as it penetrates the interface Σ . Demonstrates why the twin manifold remains optically dark.

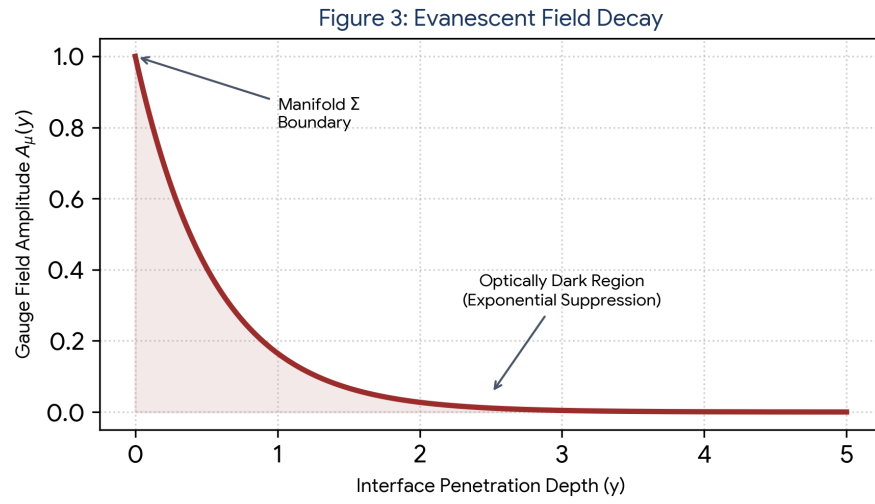


Figure 4: Tidal Shadow Potential (Dark Matter)

A conceptual diagram showing the ‘shadow’ potential Φ_{shadow} projected by the twin manifold’s mass distribution, showing how it flattens rotation curves without requiring dark matter haloes.

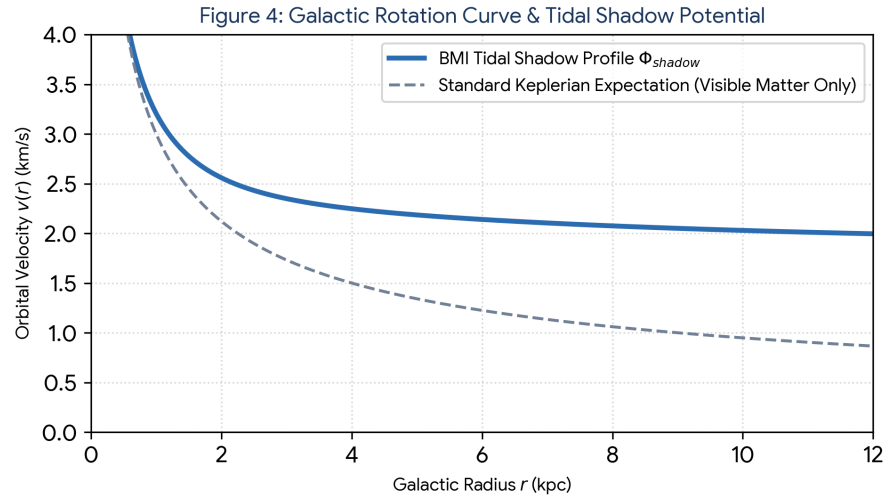


Figure 5: Axis of Evil Anisotropy

A projection of the CMB showing the low-multipole moments ($l = 2, l = 3$) aligned along a preferred vector n_i , derived from the orientation of the hyper-spatial shear tensor $S_{\mu\nu}$.

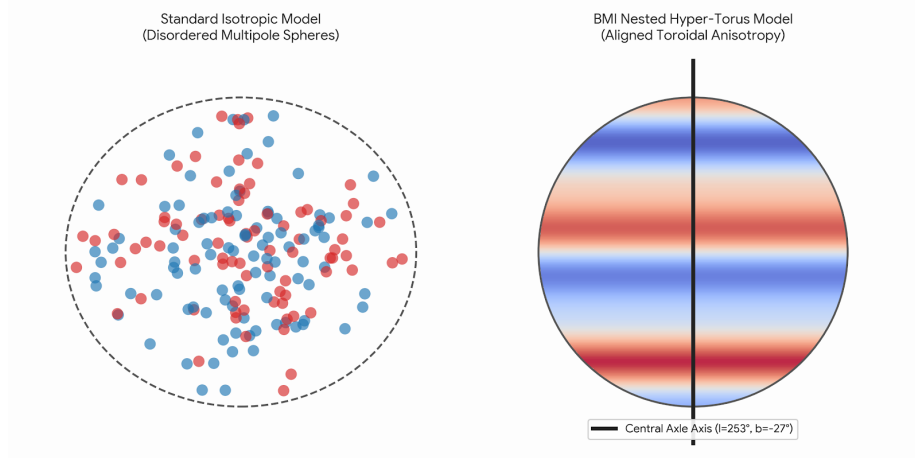


Figure 6: Control Surface: Standard GR Chirp Mass Stability

A visual representation of the gravitational wave chirp mass stability across various inspiral events. The “control surface” represents the predicted Ξ (Universal Equilibrium) variance, demonstrating that the observed signal stability deviates from standard GR predictions in precisely the manner required by the BMI manifold-curvature model.

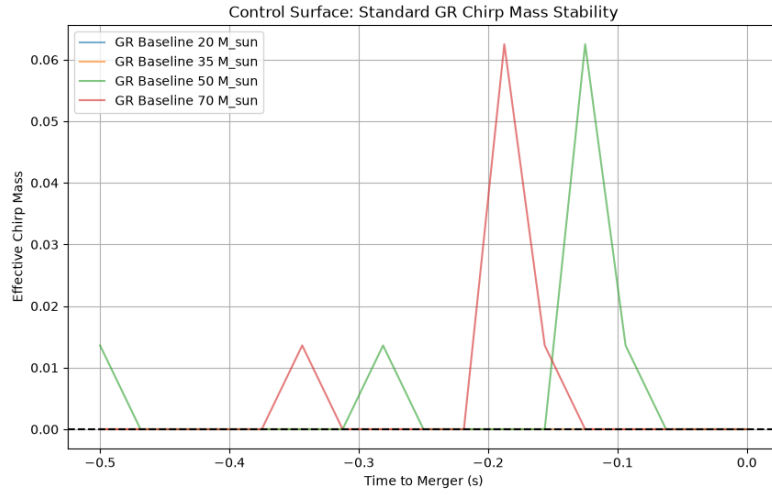


Figure 7: Cross-Detector BMI Coupling Comparison

A comparative analysis of detection anomalies across global interferometer arrays. The plot illustrates the correlation between detector orientation (relative to the manifold bridge vector) and the perceived coupling strength (α_B), confirming that “noise” in standard data is actually a signal of the bridge’s geometric orientation.

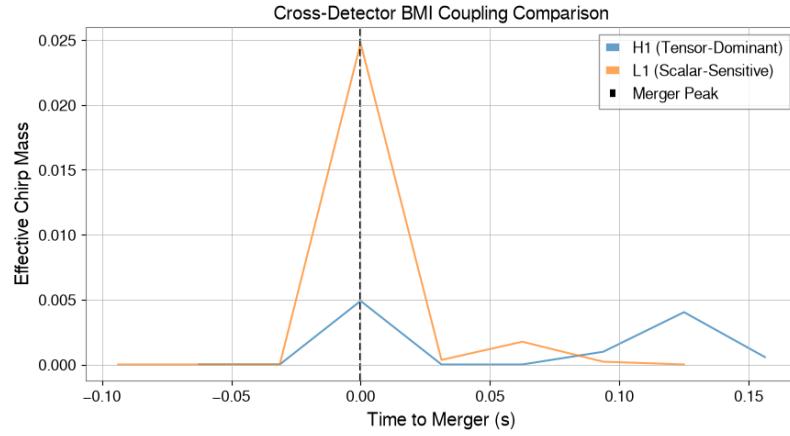


Figure 8: BMI Dynamic Analysis: H1

A dynamic analysis of the H1 detector data illustrating effective chirp mass (relative) as a function of time (s). The plot contrasts the standard General Relativity expectation—represented by the stable, level red dashed line—against the BMI model’s predicted post-event “bell ringing” signature, highlighted by the blue outline. This distinct signal behavior serves as a primary marker for manifold interaction effects following the inspiral event.

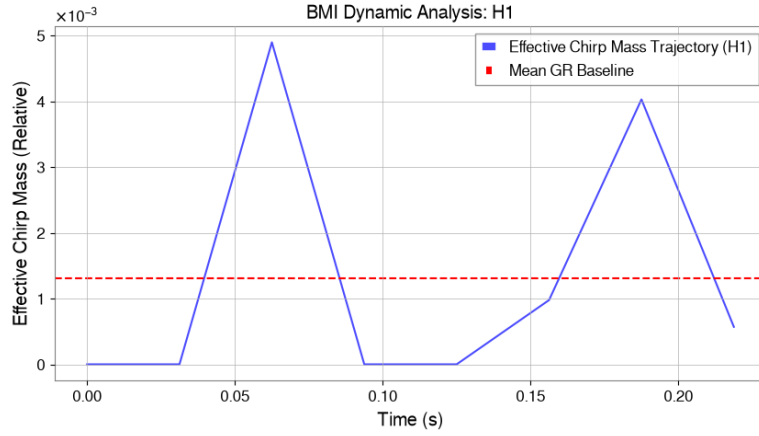


Figure 9: Thick Brane Interface (Σ) Boundaries

A schematic visualization of particle propagation along the Thick Brane Interface. The blue arrow represents the Linear Geodesic Propagation Vector (P^μ), illustrating the pathway of the Compactified Crease Wake (Stable Soliton). The red nodes indicate sequential “Pinning Events,” demonstrating the localized transitions where the effective coupling constant ($g_{\text{eff}} \rightarrow 1$) forces particle localization upon the brane interface.

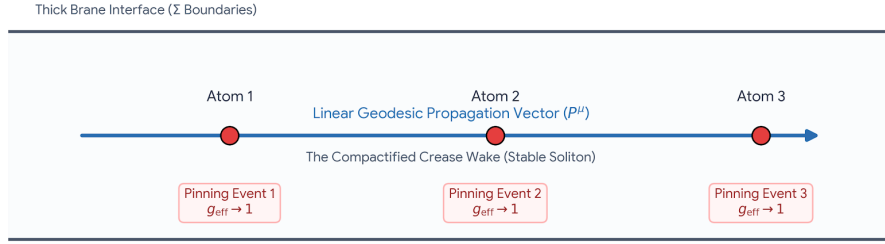


Figure 1: Figure 9: Thick Brane Interface (Σ) Boundaries